From jun-2018 | Page 28:

Q9 (b) (i): Explain why a transformer will not work with the input current as shown in Figure 17.

From jun-2018 | Page 29:

Q9 (b) (ii): A transformer has 30 turns on the primary coil and 150 turns on the secondary coil. A potential difference of 25 V is applied across the primary coil. Calculate the potential difference across the secondary coil. Use an equation selected from the list of equations at the end of this paper.

From jun-2018 | Page 30:

Q9 (c): High voltage transmission cables and transformers are used in the national grid. Explain how using high voltage transmission cables and transformers allows the distribution of electrical power around the United Kingdom to be as efficient as possible. Refer to the following equations in your answer. $P = I2 \times R$ $Vp \times Ip = Vs \times Is$

From jun-2019 | Page 17:

Q6 (b) (i): Describe the cause of the changing magnetic field in the core of the transformer.

Q6 (b) (ii): A potential difference of 230 V is applied across the primary coil of a transformer. There is a potential difference of 15 V across the secondary coil. The primary coil has 2000 turns. Calculate the number of turns in the secondary coil. Use an equation selected from the list of equations at the end of this paper.

From jun-2020 | Page 23:

Q7 (b): Explain how sound is produced when an alternating current is supplied to the coil of the loudspeaker.

From jun-2020 | Page 24:

Q7 (c) (i): State the purpose of the transformer shown in Figure 18.

Q7 (c) (ii): Calculate the output voltage of the secondary coil. Use an equation selected from the list of equations at the end of this paper.

From jun-2022 | Page 32:

Q10 (c): Using Figure 22, explain the stages in the process of delivering electricity efficiently from P to T. Your answer should include details of the effects that Q, R and S have on efficiency.

From jun-2023 | Page 32:

Q10 (b): Calculate the potential difference across the secondary coil. Use an equation selected from the list of equations at the end of the paper.

From jun-2023 | Page 33:

Q10 (c): Compare how the device operates when used as a loudspeaker with how the device operates when used as a microphone.

From june-2024 | Page 34:

Q10 (b): A transformer converts a voltage of 11 000 V to 230 V. Explain how the design of this transformer enables the voltage to be converted from 11 000 V to 230 V. Your answer should include details of the structure of a transformer; how a transformer works, using ideas of electromagnetic induction; how the design of this transformer enables this exact voltage of 230 V to be obtained.

From nov-2021 | Page 26:

Q9 (a) (i): The size of the potential difference (voltage) across the secondary coil is

Q9 (a) (ii): Explain how an alternating current in the primary coil causes an alternating current in the secondary coil of the transformer.

From nov-2021 | Page 27:

Q9 (b): The primary coil of a different transformer is connected to the 230 V mains supply. The voltage across the secondary coil is 15 V. The primary coil has 600 turns. Calculate the number of turns on the secondary coil. Use an equation selected from the list of equations at the end of the paper.